

OJAS GRAMOPADHYE

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EDUCATION

New York University, New York
Master of Science, Computer Science

Aug 2024 - May 2026
GPA : 3.93/4

Courses : Artificial Intelligence, Deep Learning, Robot Perception, Big Data, Algorithms, ML Systems & Operations

Indian Institute of Technology Bombay, India
Bachelor of Technology, Computer Science and Engineering

May 2019 - Jul 2023
GPA : 7.74/10

Courses : AI & ML, Speech Recognition, Image Processing, Database Systems, Data Structures, Software Systems

KEY PROJECTS

Flowsight : NYC Traffic Collision Analysis and Prediction

Jan 2025 - May 2025

Course Project | Big Data

New York University

- Processed millions of NYC traffic and collision records using a Kafka - MongoDB - Spark pipeline, enriching speed data with crash context through BallTree-based haversine nearest-neighbor joins
- Identified high-risk collision hotspots by training a Spark ML model, thresholding outputs into custom risk function based on incident impact, monitoring with a React dashboard and d3.js visualizations

Erica : AI Tutor agent with GraphRAG

Sept 2025 - Dec 2025

Course Project | Artificial Intelligence

New York University

- Developed a Dockerized LLM tutor agent that ingests multi-format course content, enriches it with LLM metadata, stores corpus artifacts in MongoDB, and pipelines them into GraphRAG for grounded, concept-aware retrieval
- Delivered structured, prerequisite-aware explanations by integrating semantic search, a typed knowledge graph, and Streamlit UI with subgraph visualizations, enabling the agent to answer queries with references

Multi-subject alignment : Neural Speech Decoding

Sept 2025 - Ongoing

Graduate Research | NYU Video Lab

New York University

- Improved cross-subject neural speech decoding by learning subject-invariant embeddings through contrastive alignment and adapter-based modeling across multiple ECoG participants
- Experimenting with representation learning by implementing a wav2vec2-based pretraining pipeline : masked encoder modeling, Gumbel quantization, negative sampling, and a Transformer context network for phoneme classification

WORK EXPERIENCE

Detection and Mitigation of Hallucination in Generative Models

May 2023 - May 2024

Research Assistant

IBM Research | IIT Bombay

- Improved diagnostic accuracy of LLMs to 90% by designing a Forward-Backward Chain-of-Thought evaluation pipeline and fine-tuning LLaMA-2 (7B, 70B) models for hallucination mitigation in medical QA
- Built a Subjective Medical QA benchmark and a reward-based verification pipeline that generates mixed-step reasoning signals and enables scalable, reproducible experimentation - EMNLP Findings 2024 (*arXiv: arxiv.org/abs/2403.04890*)

Japanese OCR Pipeline

May 2022 - Jul 2022

Data Science Intern

Daikin

- Delivered a full OCR pipeline for 3,000+ Japanese documents by building document-cleaning and image-normalization scripts, achieving 0.91 character-level accuracy with a MobileNetV3 + Differentiable Binarization detector
- Wrote preprocessing and evaluation pipeline to streamline dataset preparation and measure OCR performance on proprietary scanned documents

Locust Occurrence Modeling and Prediction

May 2021 - Jul 2021

Data Science Associate | R&D Team

Dtime.ai

- Supported an early-warning system for East African farmers by processing 40+ years geospatial vegetation and weather data from East African region into structured 10x10 square-km grids and handling class imbalance with SMOTE
- Improved predicted locust attack precision by 10% over baselines by performing pixel-level classification and modeling spatiotemporal patterns with a ConvLSTM U-Net model

TECHNICAL SKILLS

Machine Learning: LLMs, RAG systems, AI agents, NLP, deep learning, contrastive learning, representation learning, LoRA fine-tuning
Frameworks & Libraries: PyTorch, TensorFlow, HuggingFace, PyTorch Lightning, LangGraph, LangChain, SentenceTransformers, scikit-learn
Data & Systems: Apache Kafka, Apache Spark, Dask, MongoDB, PostgreSQL, ChromaDB, MLflow, Docker, Linux, Bash
Software Engineering: Python, C++, JavaScript, TypeScript, FastAPI, React, Streamlit, d3.js, Git, REST APIs